SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

Term 7- Sept 2024

Nanoelectronics and Technology (01.119/99.503)

Week 6 Day 1 (21-Oct 2025)

Outline

- Cleanroom activity

SINGAPORE UNIVERSITY OF
TECHNOLOGY AND DESIGN

Cleanroom Instructions

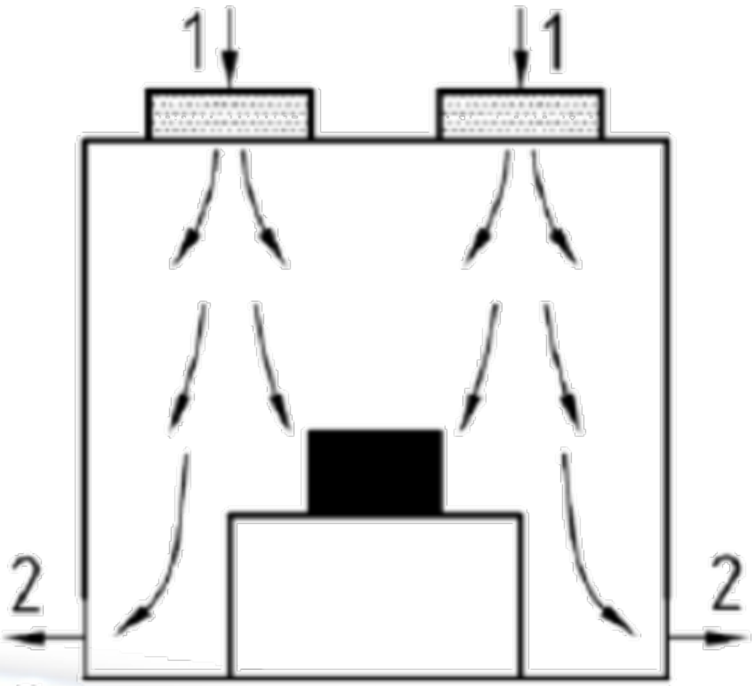
Overview

Prerequisite: ESD Control Training

1) Introduction
2) Contamination
3) Procedures
   - Before You Arrive
   - Getting Dressed
   - Entering the Cleanroom
   - Working in the Cleanroom
   - Exiting the Cleanroom

Introduction

- What is a cleanroom?
  - An environment with a **controlled** level of contamination
- How are they classified?
  - A Class (n) Cleanroom is defined as a room with air containing no more than (n) particles per cubic foot equal to or larger than 0.5μm in diameter.
    - (n) is typically 10, 100, 1,000, 10,000, 100,000, ….
    - Example: A Class 10,000 Cleanroom has less than 10,000 particles greater than or equal to 0.5μm in diameter.
    - For some reference, a human hair is 100μm in diameter!
- How does it work?
  - ressure positive room
  - Forces air out of a room
  - Inlet air is filtered



Introduction

- Why a cleanroom?
  - Contaminates, especially when conductive/metallic, can damage sensitive components
  - Particulate contaminates can end up in closed systems like propulsion lines
  - Particulate contaminates can "trick" optical navigation instruments
  - Contaminates may outgas in the vacuum conditions
    -

Contamination

- Contaminant Types

  1) Molecular Contaminants

  2) Surface Contaminants

  3) Particulate Contaminants

1) Molecular Contaminates

- Sources: Outgassing, oil vapors, alcohols, perfumes, paints, glues, epoxies
  - Aromatics: If you can smell it, suspect it as a contaminate
- Particulate filters will not handle molecular contaminates

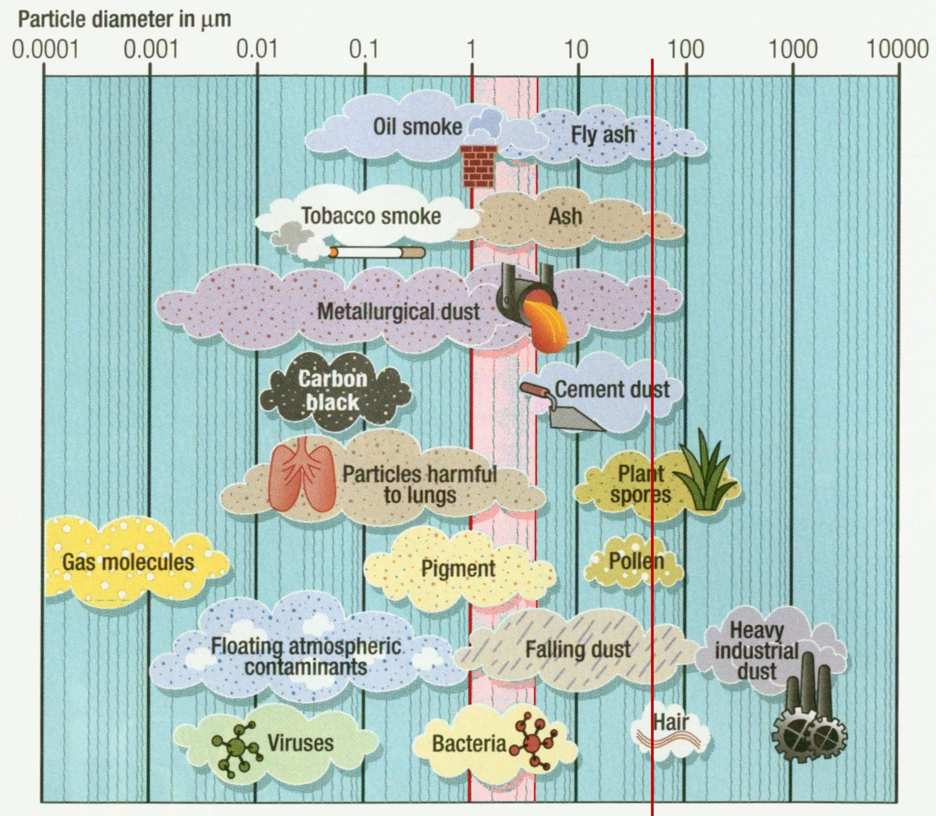
2) Surface Contaminates

- Sources: finger prints, skin oil, hand lotion, makeup
- Typically results from physical contact with hardware

Contamination

3) Particulate Contaminates

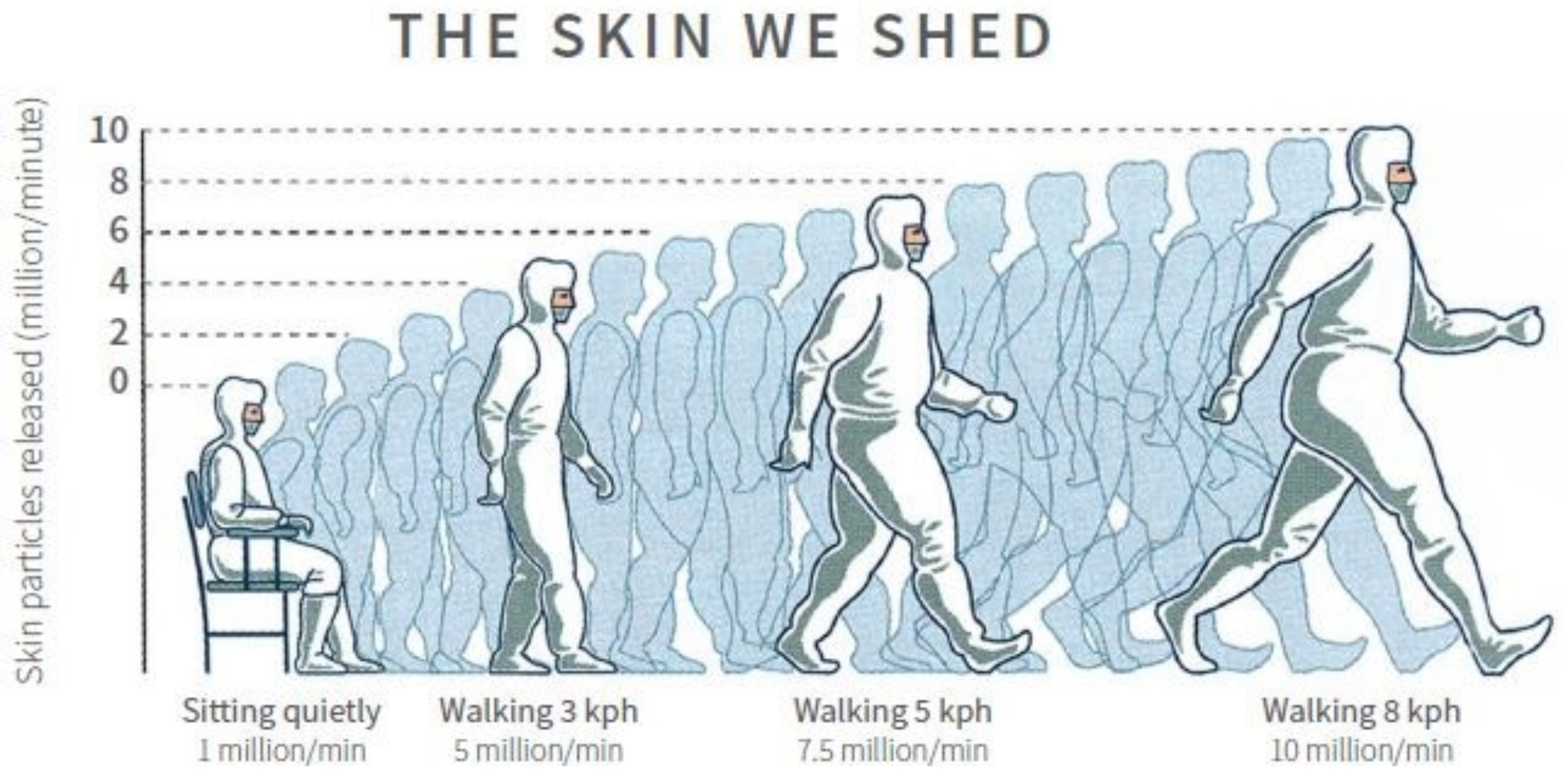
- Sources: People, particulate shedding material (cardboard, paper), abrading actions (drilling, sawing, etc…), bare wood
  - People are by far the largest source of contamination
  - Breathing, hair, skin particles (carried by convection)
- Particles are typically too small to be seen with the naked eye
  - Humans can see 50μm on an ideal background
  - Recall, we classify cleanrooms by 0.5μm particles
- Electrostatically bond to surfaces



Source: Wacker

Contamination

- People are by far the largest source of contaminates
  - Breathing
  - Coughing/Sneezing
  - Hair
  - Oils
  - Skin
    - Carried by convection



Preparing for the Cleanroom

- Before you get to B30:
  - Shower in the morning on the day you will be entering the cleanroom
  - Do not wear any makeup, hair spray, perfume, or colognes
    - Deodorant is allowed (and important)
  - Use hand lotion → Recall dry skin is an ESD risk!
- What to wear under the coat:
  - Cotton
  - Clean, closed toed shoes.
    - B30 has a shoe brush
    - Bring an extra pair on rainy, snowy, or muddy days.
  - Pants (no shorts)
  - Nothing fuzzy (sweaters) or produces lots of static
    - Nylon, wool, silk, polyester

Entering the Cleanroom

- Things to leave behind before entering the Anteroom
  - Cell phones
  - Jewelry
  - Food, drinks, and gum (not allowed in B30 anyways)
  - Pencils, erasers, cardboard, tape
- Inside the Anteroom
  - One person at a time
  - Step on the sticky mat
    - If a sticky sheet is used, **remove the mat from the Anteroom**, pull off a layer, and return it to the room
  - If you need to open a bag, use scissors
    - Do not tear anything
  - As you dress, inspect each article of clothing for damage and cleanliness
  - Discard any mask, hat, or glove that falls on the floor or is damaged

Getting Dressed

1) Hair Net

2) Beard Mask (if applicable)

3) Coat

4) Wrist Band
    - Test if not using continuous monitoring

5) Gloves

Cleanroom Procedures

- Bringing hardware into the Cleanroom:
  - Clean hardware with Isopropyl Alcohol (IPA) in the Anteroom while grounded
  - Change gloves after cleaning
- While in the Cleanroom:
  - Work in pairs!
  - Follow all ESD Protected Area Procedures
  - Do not touch any part of your exposed face
  - Move precisely and smoothly
  - Be cognizant and aware of all hardware
  - Place hardware in the center of the table and not in a location where it is susceptible to be tipped
  - Be cognizant of where your wrist strap cord is
  - Touch hardware only when necessary
  - Never tear paper or bagging material

Exiting the Cleanroom

- **Return all tools to their spot in the toolbox before exiting the clean room**
  - A disorganized clean room results in significant lost time
- Getting Undressed
  - Opposite order of getting dressed
    - Discard gloves
    - Remove wrist strap
    - Remove and hang coat
    - Discard beard mask and hat

REFERENCE:

IC Cleanroom Technology and Guidelines with Personnel Procedures

Integrated Circuit (IC) cleanrooms are specialized controlled environments where temperature, humidity, and particulate contamination are stringently maintained. Personnel behavior and discipline are critical to maintaining cleanliness since human activity contributes up to 75% of total contamination in a fab. This document summarizes cleanroom technology, operational guidelines, and detailed protocols for personnel entry, gowning, and behavior inside semiconductor facilities.

1. Cleanroom Classification
IC cleanrooms are categorized using ISO standards according to particle concentration per cubic meter of air: **ISO Class 1–3:** Photolithography, oxidation, and thin‑film deposition (≤10 particles ≥ 0.1 µm/m³). **ISO Class 4–5:** Etching, implantation, inspection, and metrology zones (**ISO Class 6–7:** Dicing, assembly, testing, and packaging areas. Each class dictates airflow rates, filter efficiency, gowning levels, and access restrictions.

2. Cleanroom Design and Airflow
Design principles focus on maintaining unidirectional airflow and positive pressure gradients: **Airflow Pattern:** Vertical laminar flow (~0.45 m/s) through HEPA or ULPA filters ensures continuous particle flushing. **Pressure Control:** Positive pressure (5–20 Pa differential) prevents infiltration from less clean areas. **Temperature & Humidity:** Maintained at 21 ± 1 °C and 45 ± 5% RH for process stability. **Lighting:** Non UV, low heat lighting; yellow lights used in photolithography zones. **Surface Materials:** Smooth, non shedding finishes (epoxy, stainless steel) to minimize particle generation. **Zoning:** Transition from gray to white areas via gowning rooms, air showers, and material pass through chambers.

3. Personnel Entry, Gowning, and Conduct
Personnel control is the most vital part of contamination prevention. Every individual entering a cleanroom must follow strict protocols: **Entry Sequence:** Remove cosmetics, jewelry, and personal items. Don shoe covers and step onto sticky mats. Pass through an **air shower** for 20–30 seconds to remove loose particles. Enter gowning room following the **clean to dirty flow** layout. **Gowning Order (ISO 5 example):** Hair cover → hood Mask → coverall suit Boot covers → gloves (double if handling wafers) **Behavior Inside:** Minimize motion and talking to reduce particle emission. No leaning or touching process tools unnecessarily. No food, drinks, or paper materials. Walk slowly; sudden movements generate turbulence. Stay within designated zones and obey directional air flow markings. **Communication:** Use intercoms, hand signals, or digital tablets instead of shouting or turning frequently. **Training:** All staff undergo periodic refresher courses on cleanroom discipline, ESD awareness, and emergency procedures. Proper human behavior reduces contamination and electrostatic discharge (ESD) events that can cause wafer defects.

4. Contamination and Equipment Control
Sources of contamination include personnel, equipment vibration, process gases, and chemicals. Control strategies: Use of **Front‑Opening Unified Pods (FOUPs)** for wafer transport to isolate wafers from the ambient environment. Equipment exhaust lines fitted with **chemical scrubbers** to remove toxic vapors. All tools and chambers undergo periodic **cleaning validation** using particle counters and molecular analyzers. Surface wipes and ultra‑pure solvents used to clean benches and cassettes daily. Grounding straps, conductive flooring, and static dissipative materials prevent **ESD damage**.

5. Environmental and Particle Monitoring
Continuous environmental monitoring ensures ISO compliance and early detection of process

contamination: Fixed sensors measure particle concentration (≥ 0.1 µm & ≥ 0.5 µm), temperature, humidity, and differential pressure. Portable counters and laser particle monitors verify local conditions near process tools. AMC sensors detect trace acids, bases, and organics that can corrode wafers. Trend data is logged and visualized through SCADA or cleanroom monitoring software for real-time alerts. Filters are validated using **DOP/PAO challenge tests** to ensure sealing integrity.

6. Maintenance and Auditing

Regular preventive maintenance keeps the cleanroom within specification: Daily: Surface wipe-downs, inspection of airlocks, pressure gauges, and gowning supplies. Weekly: Check filter frames, vibration isolation mounts, and ESD ground continuity. Monthly: Airflow balancing, duct inspection, and microbial surface sampling. Annually: Full certification against ISO 14644 standards by accredited inspectors. Non-compliance triggers corrective action and retraining of personnel involved.

7. Summary and Best Practices

Cleanroom performance depends on three pillars: **engineering control, process control, and human discipline**. Personnel adherence to gowning and behavior protocols directly influences defect rates, yield, and long-term device reliability. Continuous education, air cleanliness monitoring, and strict maintenance are indispensable for sustaining world-class semiconductor fabrication standards.